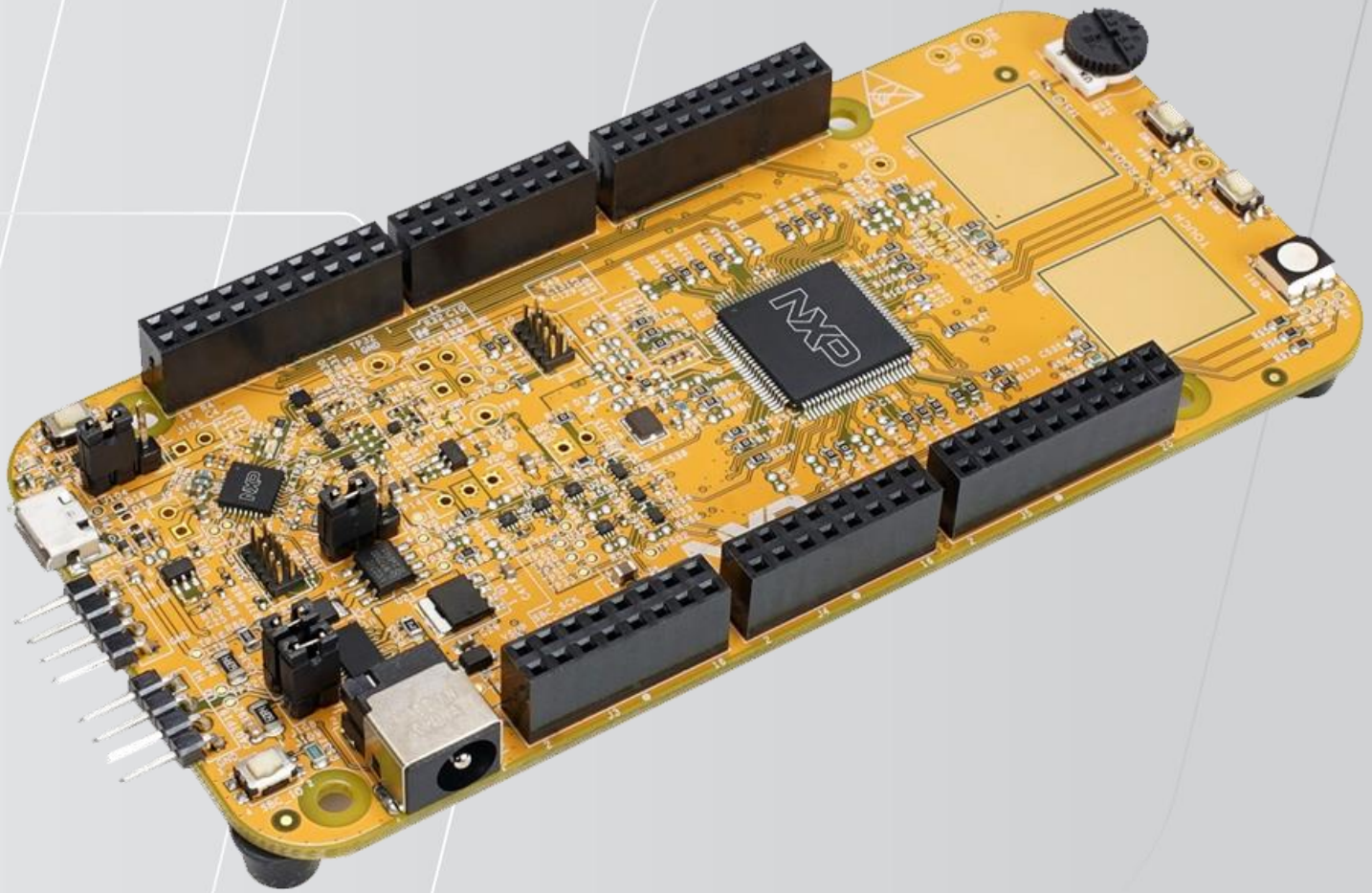
ADC
Analog-to-Digital Converter

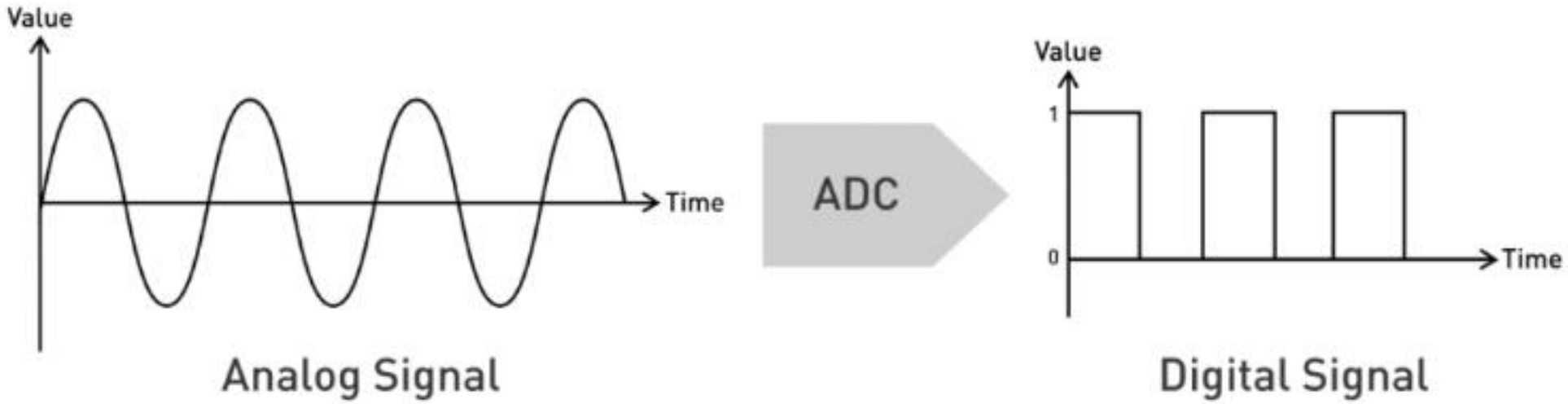


10 Oct 2024

OUTLINE

1. ADC overview
2. Analog-to-Digital Converter (ADC)
3. Configuration of ADC
4. ADC Interrupt

1. ADC overview



- **ADC** is an electronic device used to **convert analog signals** (continuous signals) into **digital signals** (discrete signals).
- Analog signals can represent any real-world parameter, such as voltage, current, temperature, or pressure.
- Digital signals are represented as binary values.

1. ADC overview

Applications of ADC:

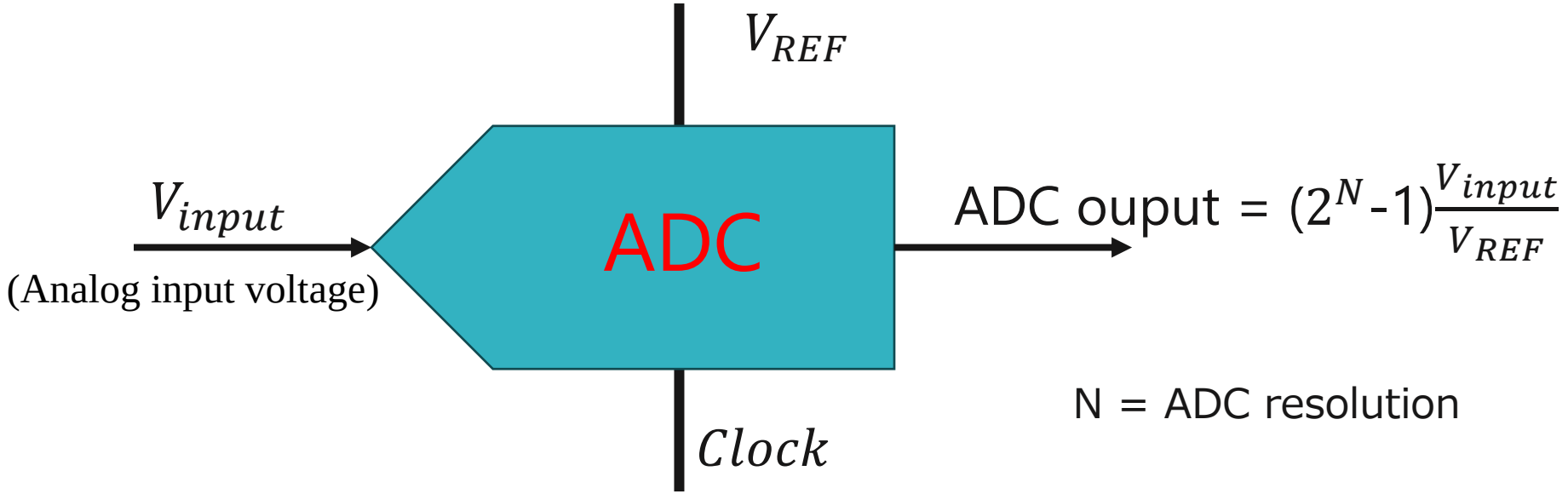
1. **Measuring Devices**: multimeters, voltmeters, and temperature sensors to convert analog signals into digital values.
2. **Data Acquisition Systems**: to convert signals from sensors and analyze them on computers.
3. **Industrial Control**: monitor analog signals from sensors, such as pressure, flow, or temperature sensors.
4. **Audio and Signal Processing**: convert analog audio signals (like from microphones) into digital signals for processing, storage, or transmission.
5. **Medical Devices**: ultrasound machines, ECGs, and other diagnostic equipment to convert biological signals into digital data for analysis.
6. **Wireless Communication**: communication devices to convert analog signals into digital, serving for transmission and encoding of information.
7. **Consumer Electronics**: digital cameras, smartphones, and smart home devices. Convert signals from sensors into digital data.
8. **Automobiles and Autonomous Systems**: sensors in cars, such as distance sensors, tire pressure sensors, and other safety systems.

1. ADC overview

ADC basic concepts:

Resolution: The number of bits in the ADC digital output representing an analog input signal.

Reference Voltage: The ADC requires a reference voltage used to create a successive approximation comparison with the analog input is compared to produce a digital output. The digital output is the ratio of the analog input with respect to this reference voltage.
VREF = VREFH – VREFL
VREFH = High reference voltage
VREFL = Low reference voltage



1. ADC overview

ADC basic concepts:

Least Significant Bits (LSB): A least significant bit (LSB) is a unit of voltage equal to the smallest resolution of the ADC, i.e. the smallest incremental voltage that causes a change in the digital output

$$\text{LSB} = \frac{V_{REF}}{2^N - 1}$$

Sampling time: the time required for the Analog-to-Digital Converter (ADC) to capture the input voltage during sampling.

Conversion Time: the time required to obtain the digital result.

$$T_{ADC} = T_{sampling} + T_{conversion}$$

Suppose ADC clock = 8 MHz, Sampling time = 13 clock cycles
For 12-bits ADC:
T_{ADC} = 13 + 12 = 25 clock cycles = 200us

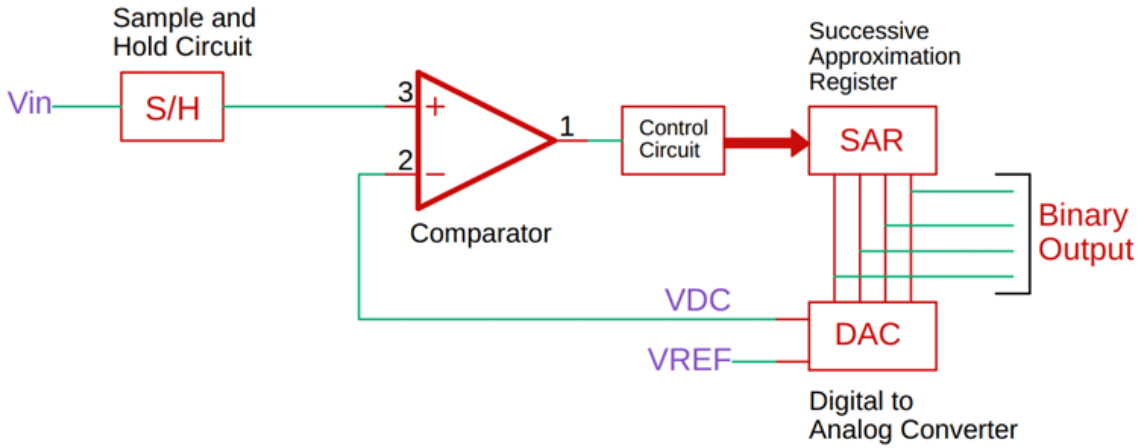
1. ADC overview

Types of ADC:

1. **Successive Approximation ADC**: Converts the analog signal to digital by comparing it to a reference voltage using a binary search method.
2. **Flash ADC**: Uses a bank of comparators to convert the analog signal to digital in a single step, making it very fast.
3. **Sigma-Delta ADC**: Works by oversampling the input signal and using noise shaping to achieve high resolution.
4. **Integrating ADC**: Measures the time it takes for an integrator to reach a certain voltage level, often used for low-frequency applications.
5. **Pipeline ADC**: Breaks the conversion process into several stages, allowing for high speed and moderate resolution.
6. **Dual Slope ADC**: Integrates the input signal and then compares it to a reference signal, commonly used in digital multimeters.

1. ADC overview

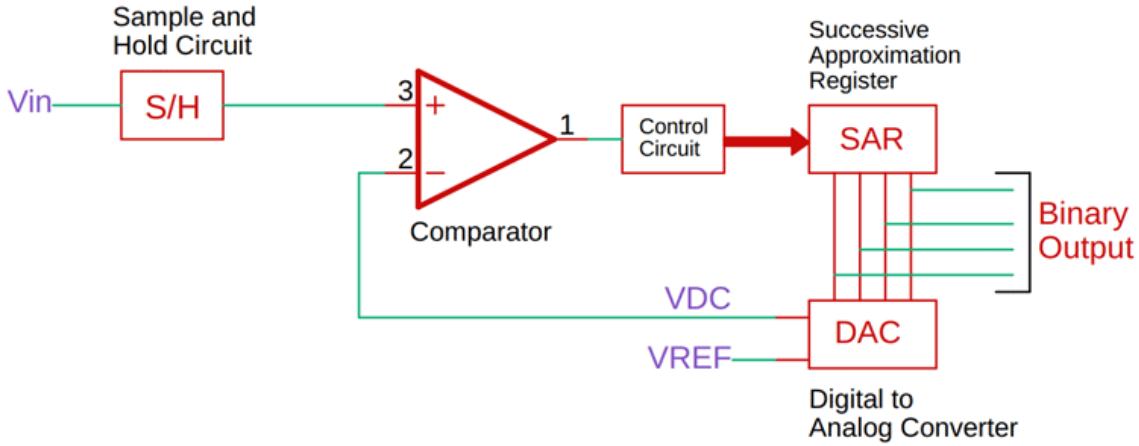
Typical Topology of a SAR ADC:



1. **Sample and Hold Circuit**: this circuit samples the analog input signal and holds it steady during the conversion process. It ensures that the ADC processes a stable voltage level.
2. **Comparator**: A high-speed comparator compares the held analog voltage to a reference voltage provided by the Digital-to-Analog Converter (DAC). It determines whether the input voltage is higher or lower than the DAC output.
3. **Successive Approximation Register (SAR)**: the SAR is a digital control unit that generates the binary search algorithm to determine the digital representation of the analog input. It stores the current approximation and updates it based on the comparator's output.
4. **Digital-to-Analog Converter (DAC)**: the DAC converts the digital output from the SAR back to an analog voltage. This voltage is then compared with the sampled input voltage.
5. **Reference Voltage Source**: A stable reference voltage is required for the DAC to provide accurate comparisons and conversions.

1. ADC overview

Operation of a SAR ADC



1. **Sampling:** The input signal is sampled and held by the sample-and-hold circuit.
2. **Initial Approximation:** The SAR sets the most significant bit (MSB) of the DAC to 1, creating an initial approximation of the input voltage.
3. **Comparison:** The comparator compares the DAC output with the sampled voltage. If the DAC output is less than the input, the SAR sets the bit to 1; if greater, it sets it to 0.
4. **Bit Shifting:** The SAR shifts down to the next bit and repeats the comparison process, adjusting the DAC output each time. This process continues until all bits are determined, typically taking N clock cycles for an N-bit resolution.
5. **Final Output:** Once all bits are processed, the final digital representation of the input analog signal is available.

2. Analog-to-Digital Converter (ADC)

The features of the ADC module:

- Linear **successive approximation** algorithm with up to 12-bit resolution
- Up to 32 single-ended external analog inputs
- Single-ended **12-bit, 10-bit, and 8-bit** output modes
- Output in **right-justified unsigned format** for single-ended
- **Single** or **continuous** conversion modes
- Automatic return to idle after single conversion
- Configurable **sample time** and conversion speed/power
- Conversion complete/hardware average complete flag and interrupt
- **Input clock** selectable from up to four sources
- Operation in **low-power** modes for lower noise
- Selectable hardware conversion trigger with hardware channel select
- Automatic **compare** with interrupt for less-than, greater-than or equal-to, within range, or out-of-range, programmable value
- **Hardware average** function
- Selectable **voltage reference**: external or alternate
- **Self-Calibration** mode

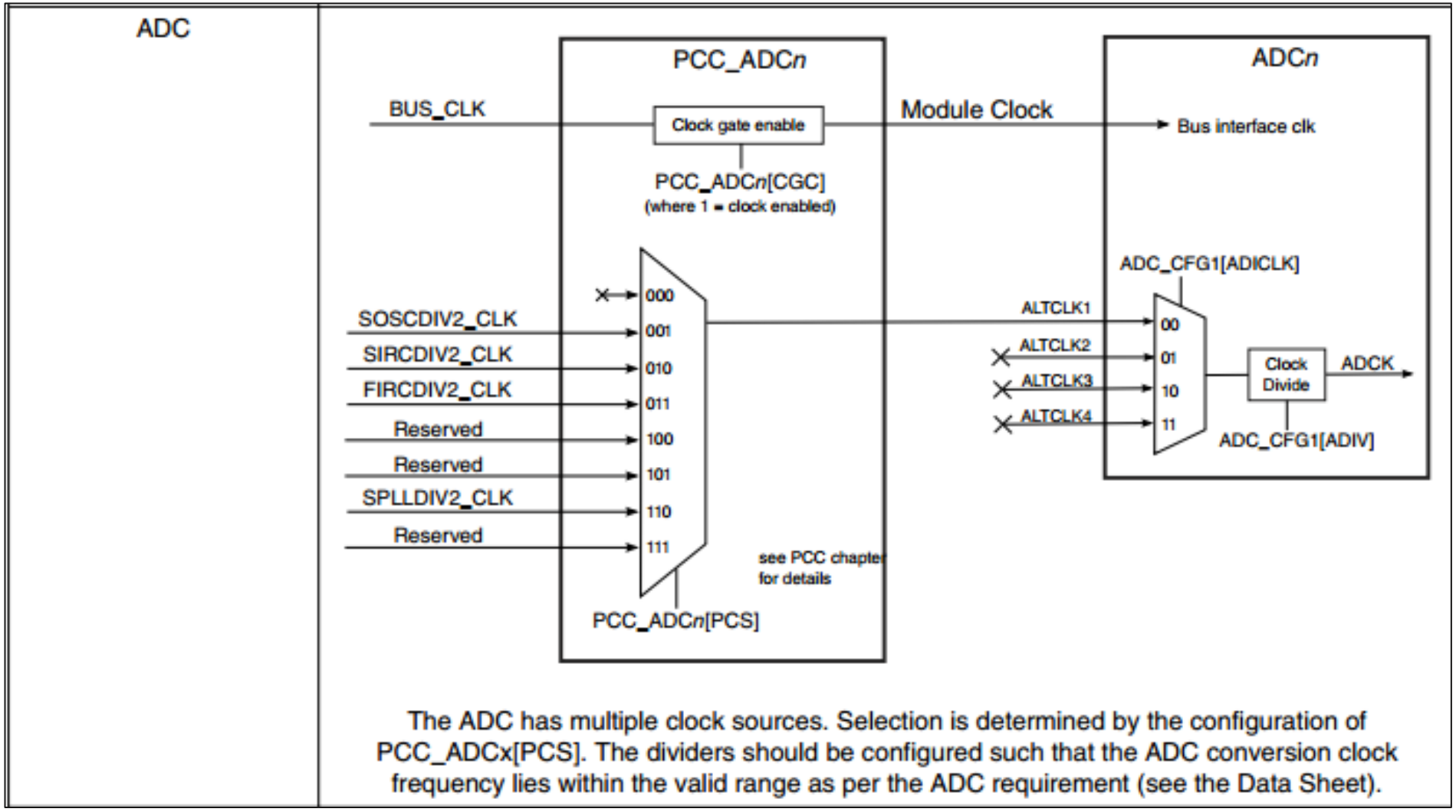
2.1. ADC signal

Table 44-1. ADC input signal descriptions

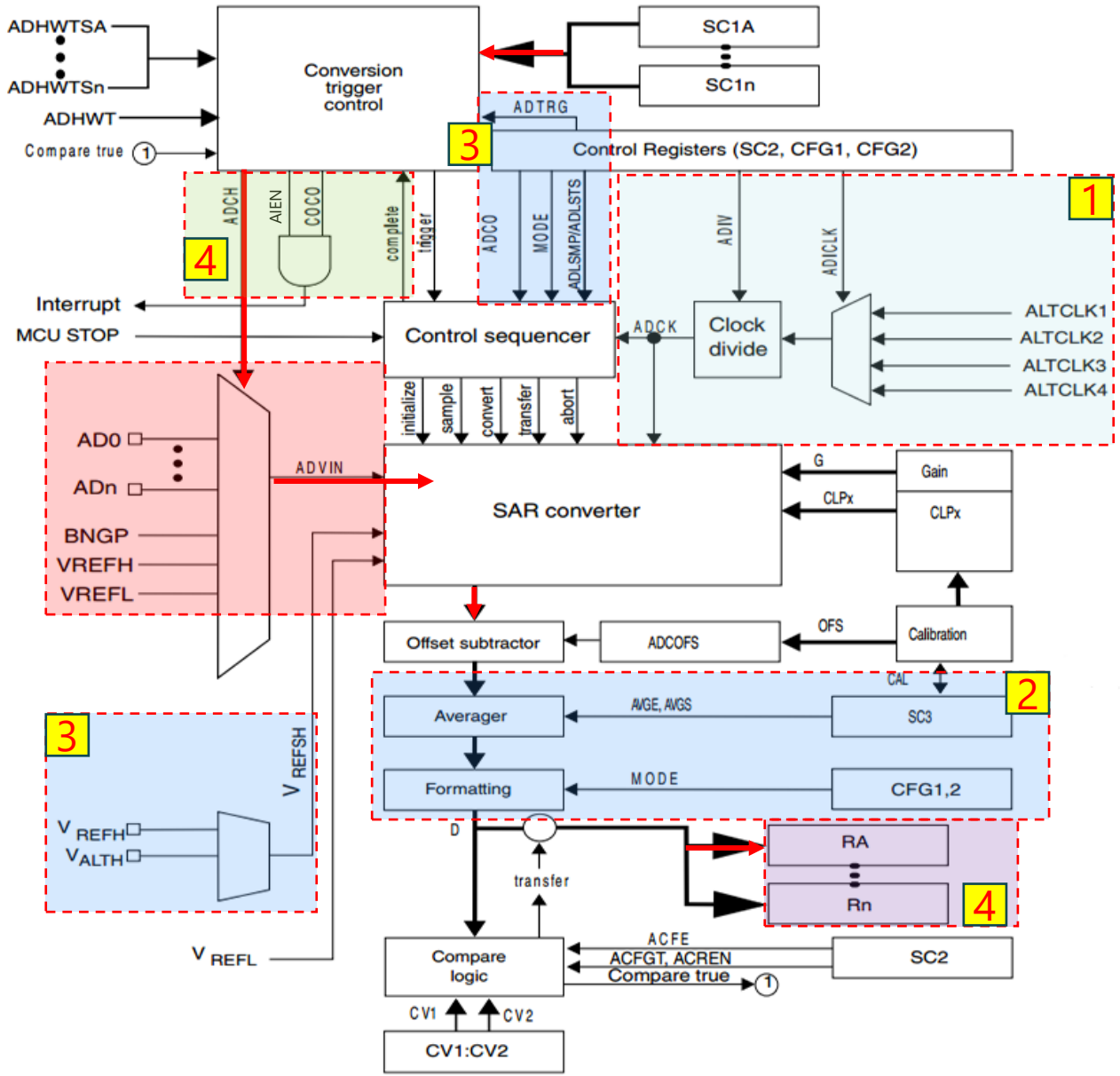
Signal	Description
AD*n*	Single-Ended Analog Channel Inputs
V_{REFSH}	Voltage Reference Select High
V_{REFSL}	Voltage Reference Select Low
V_{DDA}	Analog Power Supply
V_{SSA}	Analog Ground

- Each ADC module supports up to 32 single-ended inputs.
- The ADC analog portion uses V_{DDA} as its power connection.
- The ADC analog portion uses V_{SSA} as its ground connection.
- V_{REFSH} and V_{REFSL} are the high and low reference voltages for the ADC module.
- The ADC module supports up to 32 analog inputs. An analog input is selected for conversion through the SC1[ADCH] channel select field.

2.2. Source clock ADC

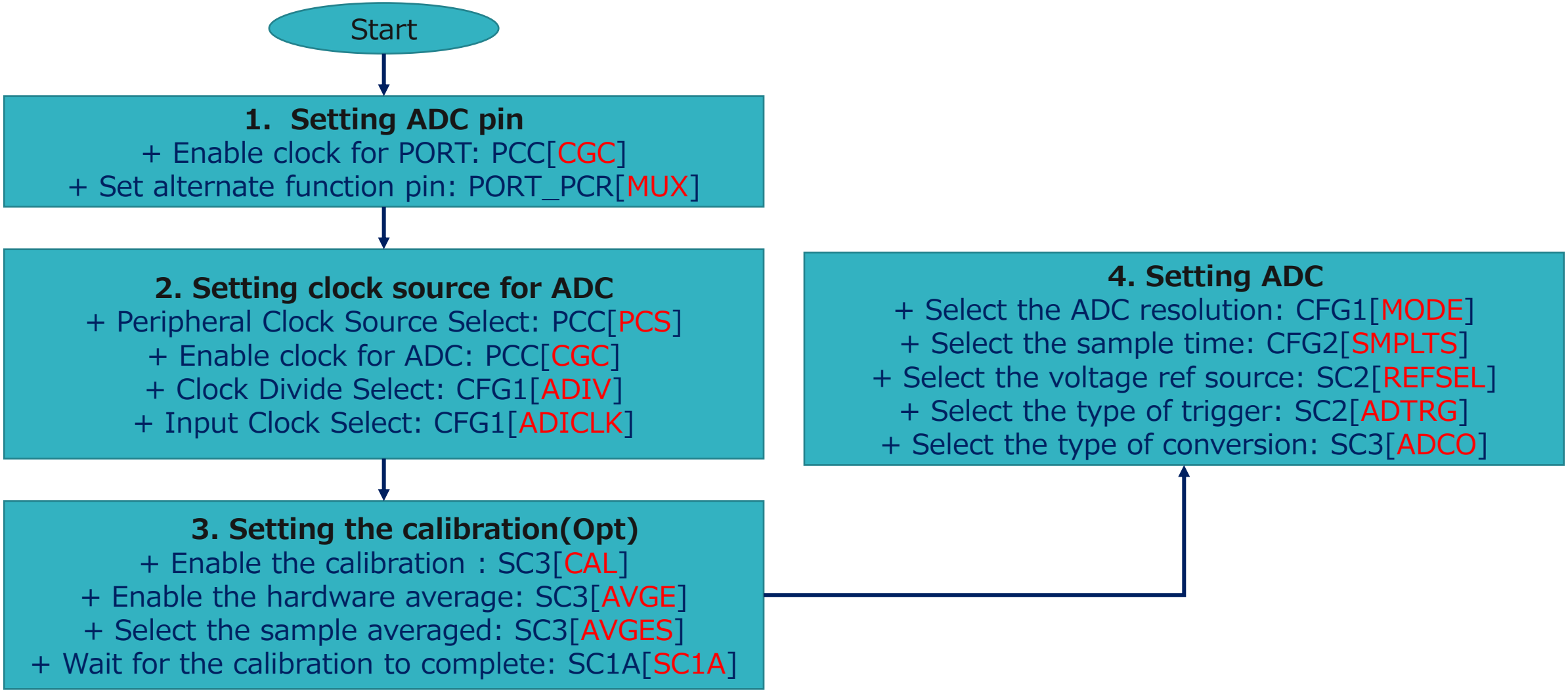


2.3. ADC module block diagram

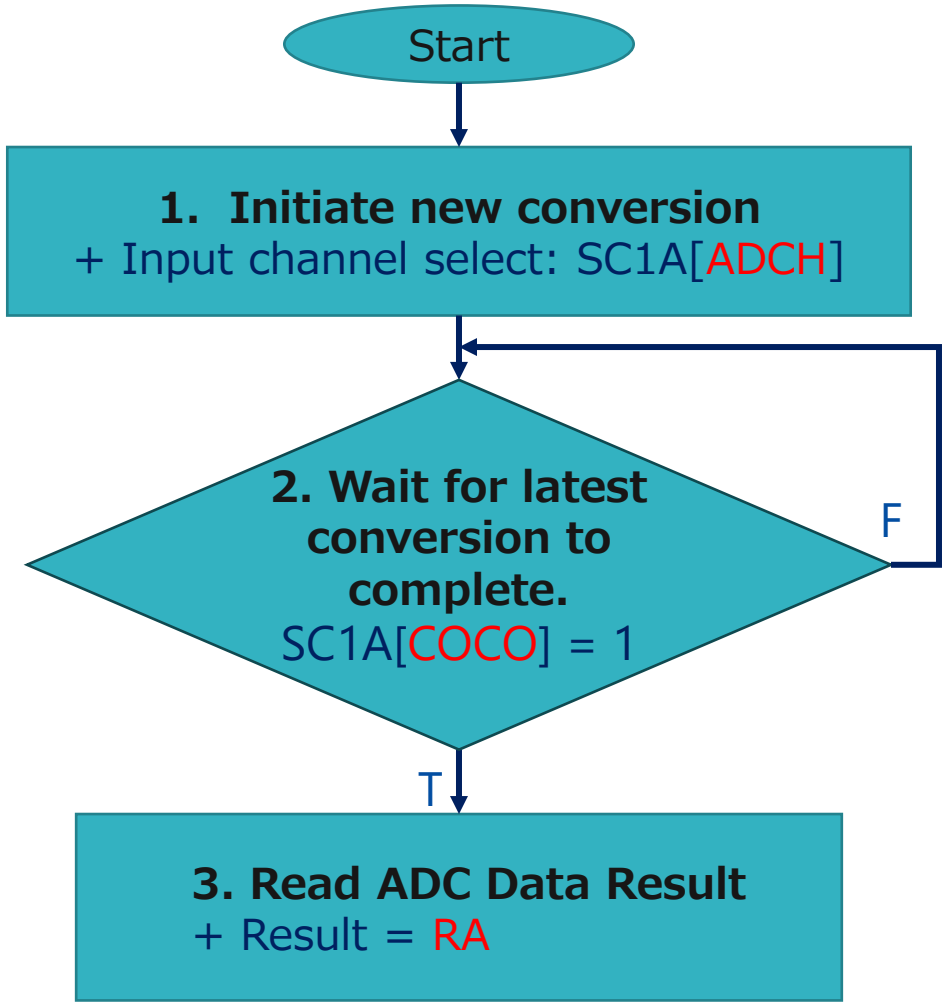


No.	Register	Field	Function
1	CFG1	ADICLK	Input Clock Select
		ADIV	Clock Divide Select
2	SC3	AVGE	Hardware Average Enable
		AVGS	Hardware Average Select
3	CFG1	MODE	Selects the ADC resolution
	CFG2	SMPLTS	Sample Time Select
	SC2	ADTRG	Conversion
		REFSEL	Voltage Reference Selection
	SC3	ADCO	Continuous Conversion Enable
4	SC1A	ADCH	Input channel select
		COCO	Conversion is completed.
		AIEN	Interrupt Enable
	RA	D	Data result

3. Configuration of ADC



3.1. ADC software triggering



3.2. Assignment 1

Write a program to **read** an analog pin:
- ADC in single conversion mode
- 12-bit resolution
- ADC software triggering
- ADC0_SE12

If the ADC result is greater than 2.5V, turn on the RED Led;
otherwise, turn off the RED Led.

3.3. Assignment 2

Write a program to **read** an analog pin:
- ADC in single conversion mode
- 12-bit resolution
- ADC software triggering
- ADC0_SE6

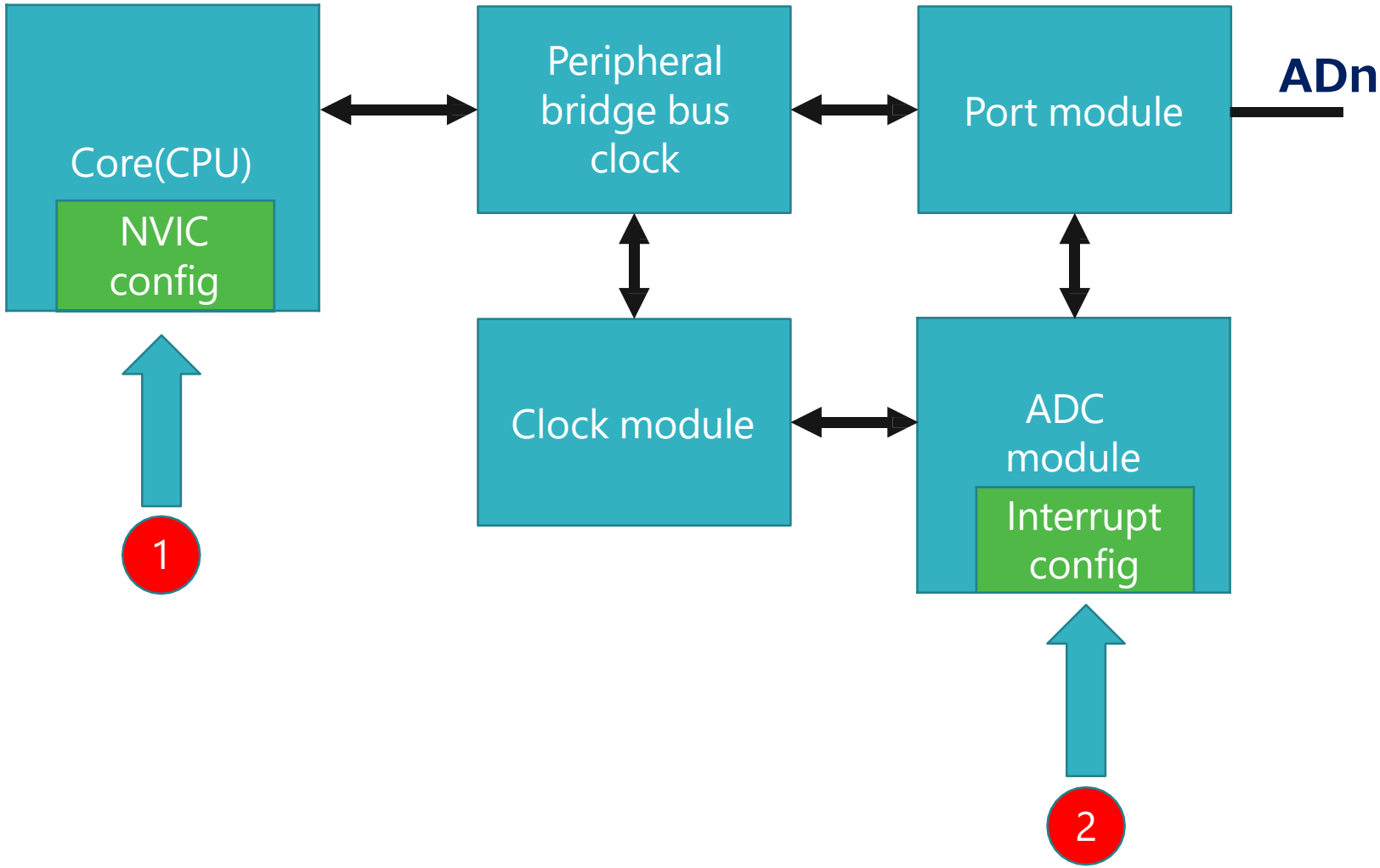
If the ADC result is greater than 2.5V, turn on the GREEN Led;
otherwise, turn off the GREEN Led.

3.4. Assignment 3

Write a program to **read** an analog pin:
- ADC in continuous conversion mode
- 12-bit resolution
- ADC software triggering
- ADC0_SE13

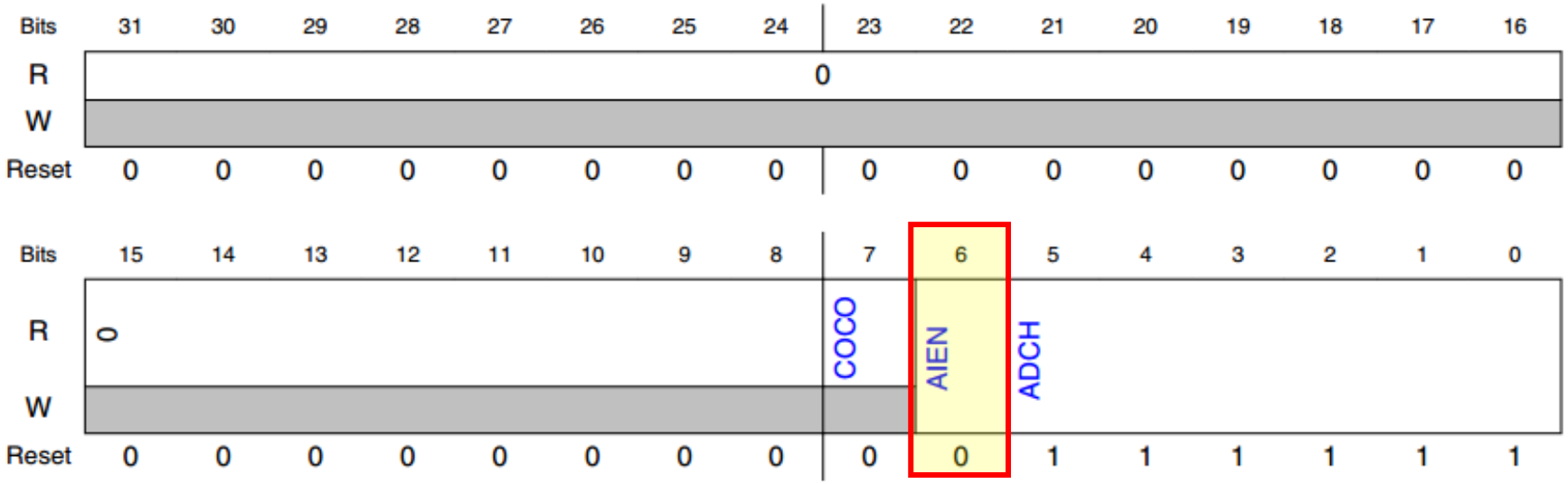
If the ADC result is greater than 2.5V, turn on the BLUE Led;
otherwise, turn off the BLUE Led.

4. ADC Interrupt



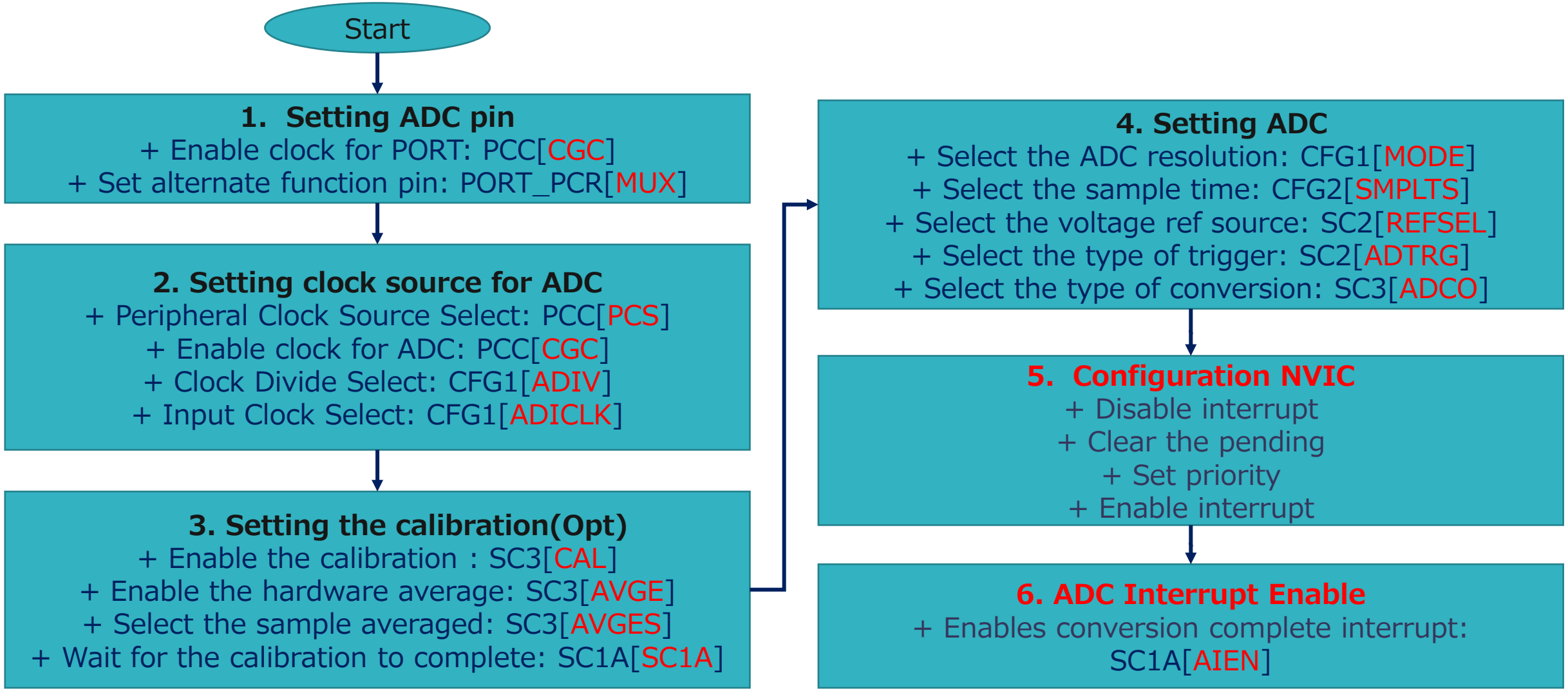
4. ADC Interrupt

ADC Status and Control Register 1 (SC1A - aSC1P):



6 AIEN	Interrupt Enable Enables conversion complete interrupts. When COCO becomes set while the respective AIEN is high, an interrupt is asserted. 0b - Conversion complete interrupt is disabled. 1b - Conversion complete interrupt is enabled.

4. ADC Interrupt



4.1. Assignment 4

Write a program to **read** an analog pin:
- ADC in single conversion mode
- 12-bit resolution
- ADC software triggering
- ADC0_SE12
- Conversion complete interrupt

If the ADC result is greater than 2.5V, turn on the BLUE Led;
otherwise, turn off the BLUE Led.

4.2. Assignment 5

Write a program to **read** an analog pin:
- ADC in single conversion mode
- 12-bit resolution
- ADC software triggering
- ADC0_SE12
- Conversion complete interrupt
- Timeout interrupt of 200ms to trigger an ADC conversion

If the ADC result is greater than 4V, turn on the BLUE Led;
else if the ADC result is greater than 3V, turn on the GREEN Led;
else if the ADC result is greater than 2V, turn on the RED Led;
otherwise, turn off all Leds.


Thank you